

1. GLOBAL ANYCAST EDGE & CLIENT LAYER

Client Browsers & Apps

Desktop React / Mobile Apps
HTTP/3, WebSockets, TLS 1.3

Cloudflare Anycast Edge CDN

DDoS Shield, Bot Management
Static Asset & Media Caching

Next.js Edge SSR / ISR Cluster

Server-Side Rendering & Revalidation
HTML Streaming & Dynamic Headers

Imgix / CloudFront Image Pipeline

On-the-fly AVIF/WebP Resizing
Responsive Image CDN Delivery

2. API GATEWAY & CORE MICROSERVICES LAYER

Kong / Envoy API Gateway

- OAuth2 / JWT Auth Validation
- Distributed Rate Limiting
- gRPC Routing & Proxy
- WAF Security Rules

Listing Service

Property metadata, photos & amenities

Reservation Lock Service

Redlock atomic booking lock engine

Dynamic Pricing Service

Demand-based seasonal pricing

User & Host Service

Profiles, verification & reviews

Messaging & Chat Service

Realtime WebSocket host-guest chat

Payment & Payout Engine

Stripe / Escrow payout processing

Review & Rating Service

Double-blind review system

Notification Engine

Email, SMS & Push notifications

3. SEARCH, GEOLOCATION & EVENT STREAMING

Elasticsearch / OpenSearch Cluster

- Uber H3 Spatial Indexing & Bounding Box Queries
- Instant date availability filtering & faceted search

Apache Kafka Event Bus & CDC

- Debezium Change Data Capture for Realtime Sync
- Event Topics: Bookings, Reviews, Payments, Logs

Redis Enterprise Distributed Cluster

- Multi-region L2 Session & Listing Cache
- Rate-limit counters & lock synchronizations

4. PERSISTENT STORAGE, MULTI-REGION DB & DEPLOYMENT INFRASTRUCTURE

CockroachDB / Multi-Region Postgres

- Active-Active Distributed SQL (US-East, EU-West, AP-South)
- ACID Compliance for Reservations & Payments
- Automated Sharding & Locality-Aware Partitioning

AWS S3 / Google Cloud Object Storage

- Multi-region replicated high-resolution property photo storage
- Automated lifecycle policies & cold archival storage
- Direct presigned URL uploads from host devices

Kubernetes (EKS/GKE) Multi-Region Deployment

- ArgoCD GitOps Continuous Deployment Pipeline
- Horizontal Pod Autoscaler (HPA) based on CPU/RPS
- Datadog APM, Prometheus & Grafana Observability